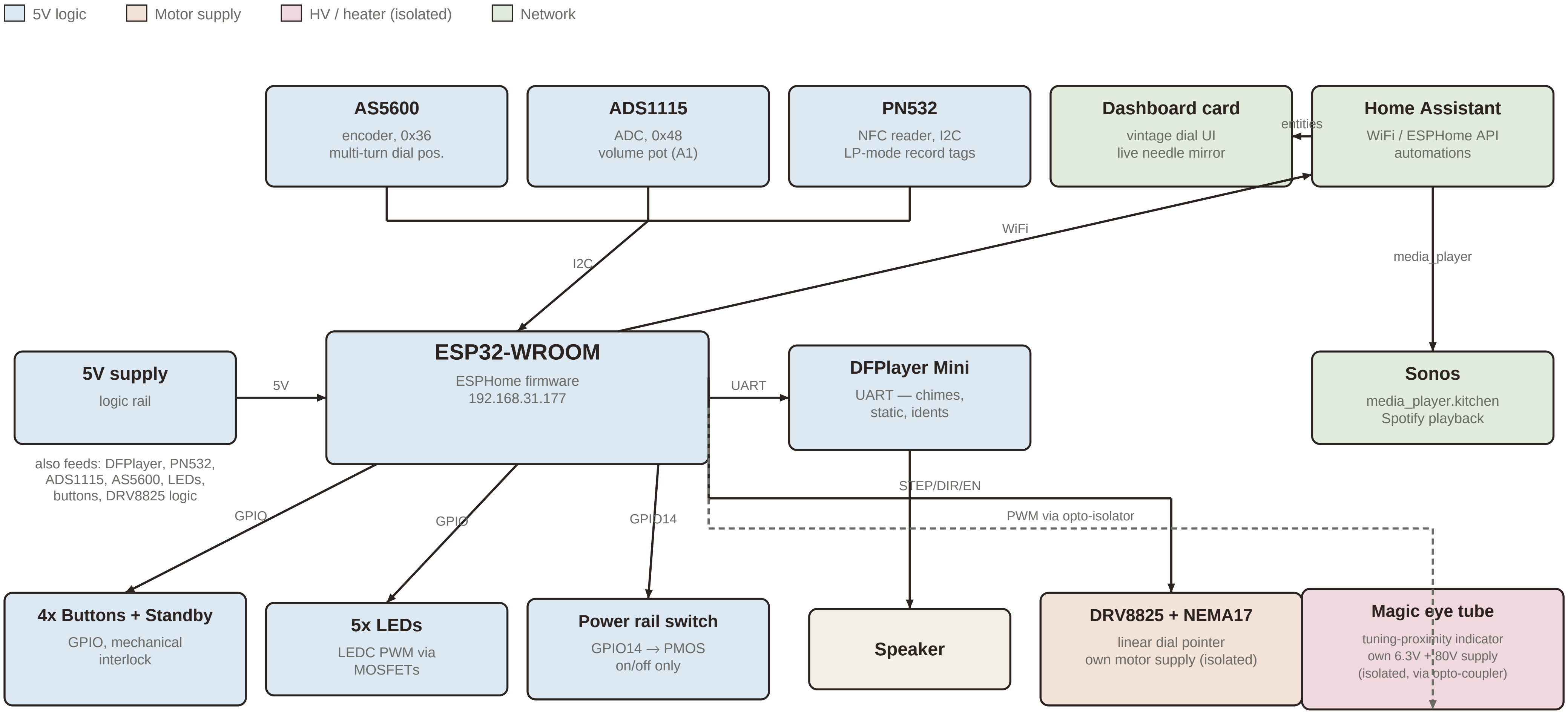
Radiola — system block diagram

ESP32 / ESPHome internet radio with motorized dial, NFC, and Home Assistant integration



Radiola — full build documentation

ESP32 / ESPHome motorized internet radio, with encoder-tracked linear dial, NFC “record” playback, DFPlayer sound effects, and Home Assistant / Sonos integration.

1. Overview

Radiola is a DIY internet radio built into a vintage chassis. Tuning works like the original set: a hand-turned knob drives a linear dial pointer along a printed band, stations sit at fixed positions, and static hisses between them. Unlike the original, the ESP32 also knows the pointer's exact position at all times (via a magnetic encoder) and can drive the pointer itself with a small stepper motor — enabling soft mechanical limits, snap-to-station, seek, a standby “parking” motion, and remote tuning from Home Assistant.

2. Subsystems

ESP32-WROOM (main controller)

Runs ESPHome firmware (radiola1.yaml). Talks to all sensors and actuators locally over I2C/UART/GPIO, and to Home Assistant over WiFi via the ESPHome native API. Static IP 192.168.31.177.

AS5600 magnetic rotary encoder (I2C, 0x36)

Absolute single-turn angle sensor reading a magnet on the dial mechanism. Polled every 50 ms; wraparound between samples is accumulated into a multi-turn position (persisted to flash) so the firmware always knows exactly where the pointer is, even across power cycles and standby. Must be powered from the always-on logic rail, not any switched supply, so tracking continues in standby.

ADS1115 ADC (I2C, 0x48)

Reads the volume potentiometer on channel A1. The former dial potentiometer on A0 was retired once the AS5600 took over tuning — channel A0 is currently free.

PN532 NFC reader (I2C)

Detects tagged “records” placed on the radio. In LP mode, a recognized tag triggers album playback on the Sonos via a Home Assistant service call.

DFPlayer Mini (UART)

Plays local WAV files from a microSD card: startup/shutdown chimes, station-tuning static (volume-scaled to the knob and to distance-from-station), and seek sounds. Actual music playback happens on a Sonos speaker via Home Assistant, not through this module.

DRV8825 stepper driver + NEMA17 motor (linear dial)

Drives the physical pointer along the dial cord. EN is held low (driver disabled, knob free-spinning) during normal hand tuning; the firmware energizes it only near the calibrated soft endstops (spring-back — assists turning back in range, resists going further out, never locks the knob) or during a commanded move (seek, snap-to-station, standby park, LP progress). Powered from its own dedicated motor supply, isolated from the logic rail, since stepper current transients would otherwise disturb the 5V rail.

5x status LEDs (LEDC + MOSFET outputs)

Dimmable via PWM. Used for the startup/shutdown “wake-up” sweep animation and as calibration feedback (blinking during the limit-setting sequence).

4x latching buttons + standby contact plate (GPIO)

Mechanically interlocked piano-key buttons (pressing one releases any other). Button functions: mode select (radio/LP), seek, and the 5-press calibration trigger. A separate spring contact plate detects the physical standby position and cuts the power rail via a GPIO-driven switch (converted from an earlier dimmable-light implementation, since PWMing a power rail risked chopping the DFPlayer's supply).

Magic eye tuning indicator (optional add-on)

A vacuum “magic eye” tube whose shadow angle narrows as the dial approaches a station, driven from the same tuning-distance signal used for the static “swim” effect. Requires its own supply: a 6.3V heater winding/regulator and an 80V HV boost for the anode/grid. The control signal from the ESP32 GPIO must pass through an optocoupler (or at minimum a series resistor + clamp) so an HV-side fault cannot reach the logic board. The HV boost converter should be isolated from the shared 5V rail, both to avoid loading it and to keep switching noise away from the I2C bus and encoder.

Home Assistant + Sonos + dashboard card

HA automations watch the ESP32's published sensors (Dial Mode, Radiola Preload Station, LP Progress) to start/stop Sonos playback, preload the next station's playlist muted while the dial is still in the static gap, and ramp volume on arrival. A custom Lovelace card renders a vintage dial face with a live needle mirroring the physical pointer, a mode badge, station/track label, and play/pause control.

3. Power budget (5V logic rail)

Estimated current draw for everything sharing the 5V logic supply. The NEMA17 motor and the magic eye's heater/HV supply are deliberately kept off this rail (see Subsystems).

Component	Typical	Peak	Notes
ESP32-WROOM	~80 mA	500 mA	WiFi TX bursts spike hard and briefly
DFPlayer Mini	40–60 mA	300–500 mA	Volume- and speaker-load dependent
PN532	~20 mA	100–150 mA	Peaks only while energizing a tag
5x LEDs	20–40 mA	75–100 mA	All five lit (startup sweep) is the peak case
AS5600	6.5 mA	~7 mA	Always on, incl. in standby
DRV8825 logic (VDD only)	4 mA	10 mA	Coil current is on the separate motor supply
ADS1115	0.15 mA	0.26 mA	Negligible
Pot / pull-ups / misc	~1–2 mA	~2 mA	Negligible
Total	**~190 mA**	**~1.2 A**	**≈ 1 W typical / ≈ 6 W peak — supply: 5V/2A min**

If a magic eye tube is added, budget an additional ~580–650 mA if its heater/HV boost shares the 5V rail (not recommended — give it a dedicated supply instead, both for current headroom and to keep switching noise off the I2C bus).

4. Firmware behavior summary

- **Soft endstops:** dial_min/dial_max calibrated by hand; the motor springs the pointer back toward the safe span near either limit rather than locking it in place.
- **Stations:** four positions at 20/40/60/80% of the usable dial span, each with a capture zone.
- **Snap-to-station:** once the hand-turned dial settles near a station, the motor centers it.
- **Seek:** sweeps to the next station with static hissing until arrival.
- **Static “swim”:** hiss volume scales with distance from the nearest station and the volume knob.
- **Preload:** while tuning through static, a sensor names the station being approached (direction-aware, with jitter hysteresis) so Home Assistant can start that playlist muted before arrival.
- **Standby park:** the pointer glides to the left end while the shutdown chime plays, then the power rail cuts.
- **Calibration:** 5x press of Button 4 enters calibration; Button 3 captures the left then right physical limits.
- **Failsafe:** if the encoder goes silent for more than 1 second, the motor is disabled and moves are locked out until recalibrated.

5. Key GPIO / bus reference

Signal	Interface	Notes
I2C bus (SDA/SCL)	I2C, shared	AS5600 (0x36), ADS1115 (0x48), PN532
UART TX/RX	UART	DFPlayer Mini
DRV8825 STEP / DIR / EN	GPIO	EN active-low; substitution placeholders — set to actual wiring
Power rail switch	GPIO14	Drives PMOS gate, on/off only — never PWM
5x LED outputs	LEDC (PWM)	Via MOSFETs
4x buttons	GPIO, INPUT_PULLUP	Mechanically interlocked
Standby contact plate	GPIO4	Plate on GND = active; away = standby
Magic eye control	GPIO via opto-isolator	Add-on — isolate from logic rail

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Source: esphome-devices/radiola (radiola1.yaml, README.md, homeassistant/). Values marked as estimates should be verified against real hardware.